

Inferring Strong Gravitational Lens Parameters from Images with Simulation-Based Inference

Strong Lens Inference Project

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Abstract

We infer the physical parameters of strong gravitational lens systems directly from single noisy images using amortized simulation-based inference (SBI). A forward model built with `lenstronomy` simulates 64×64 -pixel images of a singular isothermal ellipsoid (SIE) lens with external shear magnifying a Sérsic source, including PSF convolution and realistic Gaussian and Poisson noise. A BayesFlow neural posterior estimator—a convolutional summary network coupled to a conditional normalizing flow—is trained on 8,000 simulated image–parameter pairs and evaluated on 300 held-out systems. The Einstein radius, source position, and source size are recovered accurately ($R^2 = 0.88\text{--}0.95$), the lens ellipticity moderately ($R^2 \approx 0.5$), while external shear remains largely unconstrained at the signal-to-noise ratio of our images, consistent with physical expectations. Simulation-based calibration reveals mild overconfidence of the learned posteriors for the best-constrained parameters. Finally, extending the parameter vector with an external convergence κ reproduces the classical mass-sheet degeneracy: the κ posterior contracts only 12% relative to the prior while exhibiting a strong anti-correlation with the Einstein radius ($\langle \text{corr}(\kappa, \theta_E) \rangle = -0.83$). Training the model takes under twenty minutes on a consumer GPU (NVIDIA RTX 3060), after which posterior sampling for a new lens image takes milliseconds.

1 Introduction

Strong gravitational lensing occurs when a massive foreground galaxy bends the light of a distant background source, producing arcs, multiple images, or complete Einstein rings. The observed image morphology encodes the projected mass distribution of the lens: the ring radius traces the total enclosed mass (via the Einstein radius θ_E), the arc shapes trace the lens ellipticity, and subtle distortions trace the tidal field of neighbouring structures (external shear). Extracting these parameters from images is a core task in astrophysical applications ranging from measuring dark-matter substructure to time-delay cosmography.

Classical lens modelling fits a parametric model to each image with likelihood-based sampling (e.g. MCMC), which is accurate but slow—often hours per system—and must be repeated from scratch for every new image. Upcoming surveys (Euclid, LSST) are expected to discover $10^4\text{--}10^5$ new lenses, making per-system MCMC impractical. *Amortized* simulation-based inference offers an alternative: a neural network is trained once on simulated (parameter, image) pairs to approximate the posterior $p(\boldsymbol{\theta} \mid \mathbf{x})$, after which inference for any new image requires only a forward pass. We implement this approach with BayesFlow [1, 2], using `lenstronomy` [3] as the simulator.

Two questions structure the analysis:

1. How well can each lens and source parameter be recovered from a single noisy image, and are the inferred uncertainties statistically calibrated?
2. When an external convergence κ (a uniform mass sheet) is added to the model, does the inference correctly reflect the well-known *mass-sheet degeneracy*—i.e. report that κ is essentially unidentifiable from a single image rather than hallucinating a tight constraint?

2 Forward Model and Simulated Data

2.1 Lens and source model

Each system consists of a singular isothermal ellipsoid (SIE) lens with external shear, magnifying a Sérsic source. The inferred parameter vector is

$$\boldsymbol{\theta} = (\theta_E, e_1, e_2, \gamma_1, \gamma_2, x_s, y_s, R_s) \quad [+ \kappa], \quad (1)$$

where θ_E is the Einstein radius, (e_1, e_2) the lens ellipticity components, (γ_1, γ_2) the external shear components, (x_s, y_s) the source position, R_s the source half-light radius, and κ the optional external convergence (Section 4.4). Following the assignment specification, nuisance quantities are fixed and assumed known: the lens centroid (measurable from the lens light), the source amplitude, the source Sérsic index ($n = 2$), and the source-light ellipticity.

2.2 Priors

Parameters are drawn from independent uniform priors (Table 1). Ellipticity is sampled in axis ratio q and position angle ϕ and converted to (e_1, e_2) ; shear is sampled in modulus γ_{ext} and angle ϕ_{ext} and converted to (γ_1, γ_2) . Identical priors are used at train and test time, a prerequisite for the calibration test in Section 4.2 to be valid.

Table 1: Prior ranges. Lengths in arcsec, angles in radians.

Parameter	Range	Meaning
θ_E	(0.7, 1.6)	Einstein radius
q	(0.6, 1.0)	lens axis ratio $\rightarrow (e_1, e_2)$
ϕ	(0, π)	lens position angle
γ_{ext}	(0, 0.08)	external shear modulus $\rightarrow (\gamma_1, \gamma_2)$
ϕ_{ext}	(0, π)	external shear angle
x_s, y_s	(−0.3, 0.3)	source position
R_s	(0.05, 0.25)	source half-light radius
κ	(0, 0.20)	external convergence (only in the 9-parameter run)

2.3 Image simulation and noise

Images are rendered on a 64×64 grid at $0.05''/\text{pixel}$ (a $3.2''$ field of view), convolved with a Gaussian PSF of FWHM $0.1''$, and degraded with Gaussian background noise ($\sigma_{\text{bkg}} = 0.1$ per pixel) plus Poisson shot noise for an effective exposure time of 100 s, using `lenstronomy`’s noise utilities. With the adopted source amplitude the realized peak signal-to-noise ratio of the dataset has median ≈ 4 (range $\approx 1\text{--}8$), somewhat below the nominal 10–30 target; the images are therefore on the noisy end of realistic, which should be kept in mind when interpreting the recovery results (Section 5).

We precompute $N_{\text{train}} = 8,000$ training, $N_{\text{val}} = 1,000$ validation, and $N_{\text{test}} = 300$ held-out test systems. Simulation is CPU-bound and parallelized over worker processes; generation of the full dataset takes a few minutes on a modern desktop.

3 Inference Method

3.1 Neural posterior estimation with BayesFlow

We use BayesFlow’s neural posterior estimation: an *amortized* approximation $q_\phi(\boldsymbol{\theta} \mid \mathbf{x})$ to the posterior is trained by maximizing the log-density of true parameters under the model across simulated pairs $(\boldsymbol{\theta}, \mathbf{x}) \sim p(\boldsymbol{\theta})p(\mathbf{x} \mid \boldsymbol{\theta})$. Two networks are learned jointly:

Summary network. A compact convolutional network compresses each $64 \times 64 \times 1$ image into a 48-dimensional learned summary vector: three convolutional blocks (two 3×3 convolutions each with 32, 64, and 128 filters, followed by max-pooling and batch normalization), global average pooling, and two dense layers. BayesFlow’s built-in summary networks target sets and time series, so this custom CNN is supplied for image data.

Inference network. A conditional coupling-flow normalizing flow (depth 4) maps a standard Gaussian base distribution to the posterior over θ , conditioned on the image summary. After training, posterior sampling for a new image is a single forward pass—no MCMC.

The target parameters are standardized (z-scored) before entering the flow, since they span very different numeric scales ($\theta_E \sim 1$ vs. $\gamma_i \sim 10^{-2}$).

3.2 Training

The approximator is trained offline on the precomputed dataset for 40 epochs with batch size 32 and learning rate 10^{-3} (Keras 3 with the PyTorch backend). On an NVIDIA GeForce RTX 3060 the full training run completes in under 20 minutes; the same configuration required ≈ 1.4 hours on CPU, a speedup of roughly $4\text{--}5\times$ end to end. Figure 1 shows the loss (negative log-posterior density): the validation loss drops steeply within the first three epochs and plateaus near 4.5–5.5 after epoch ~ 17 , while the training loss continues to decrease towards ~ 0.1 —clear evidence of overfitting in the later epochs. Early stopping around epoch 20 would likely yield an equally good or better model in half the time (Section 5).

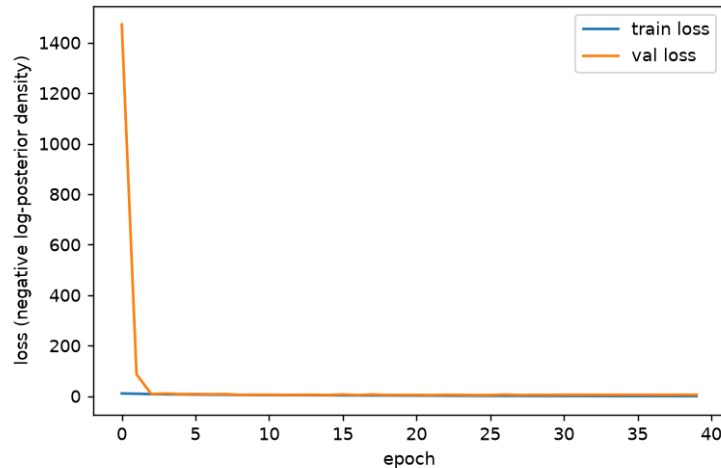


Figure 1: Training and validation loss (negative log-posterior density). The validation loss plateaus early while the training loss keeps decreasing, indicating overfitting beyond roughly epoch 20.

4 Results

4.1 Parameter recovery

Figure 2 compares the posterior mean against the true value for each parameter across the 300 test systems (2,000 posterior samples per system); Table 2 summarizes the coefficient of determination R^2 .

Table 2: Recovery accuracy (posterior mean vs. truth) on the 300-system test set.

Parameter	R^2	Assessment
R_s	0.95	excellent
θ_E	0.88	very good
y_s	0.84	very good
x_s	0.83	very good
e_2	0.49	moderate
e_1	0.47	moderate
γ_1	0.21	poor
γ_2	0.06	essentially unconstrained

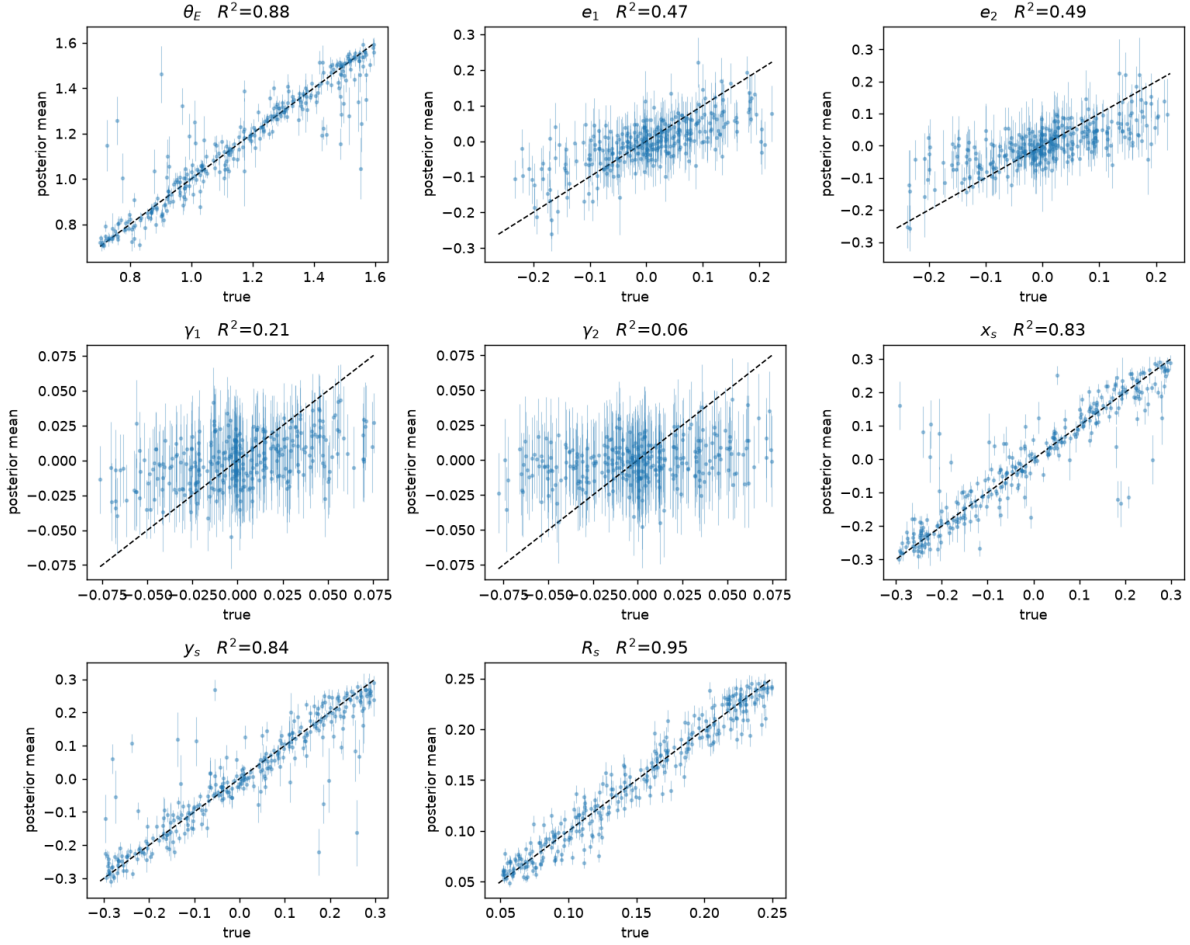


Figure 2: Posterior mean vs. true value for each parameter over the held-out test set. Error bars show the posterior standard deviation; the dashed line marks perfect recovery.

The hierarchy of recovery quality follows directly from how strongly each parameter imprints on the image:

- **Einstein radius, source position and size** (θ_E , x_s , y_s , R_s): these set the ring radius, the arc configuration, and the arc thickness—large, direct morphological signatures—and are recovered with $R^2 \geq 0.83$. R_s is the best-recovered parameter ($R^2 = 0.95$).
- **Lens ellipticity** (e_1 , e_2): moderate recovery ($R^2 \approx 0.5$) with visible shrinkage of extreme true values toward zero. This regression-to-the-prior-mean behaviour is expected

for weakly identified parameters: where the likelihood is uninformative the posterior mean reverts toward the prior centre.

- **External shear** (γ_1, γ_2): poorly constrained ($R^2 = 0.21$ and 0.06). This is physically reasonable rather than a network failure. The prior restricts the shear modulus to ≤ 0.08 , whose subtle quadrupole distortion of the arcs is comparable to, and partially degenerate with, the effect of lens ellipticity—and it is easily buried at the median peak SNR of ~ 4 of these images. The wide error bars in Figure 2 show the posterior itself “knows” shear is uncertain, which is exactly the desired behaviour of a well-behaved posterior (contrast with the calibration caveats below).

4.2 Calibration (simulation-based calibration)

Accuracy of the posterior mean is not sufficient; the posterior *width* must also be statistically correct. We use simulation-based calibration (SBC) [4]: for each test system the rank of the true parameter within the posterior samples is computed; if the posterior is calibrated, ranks are uniform, so the histograms in Figure 3 should be flat within the grey 99% band.

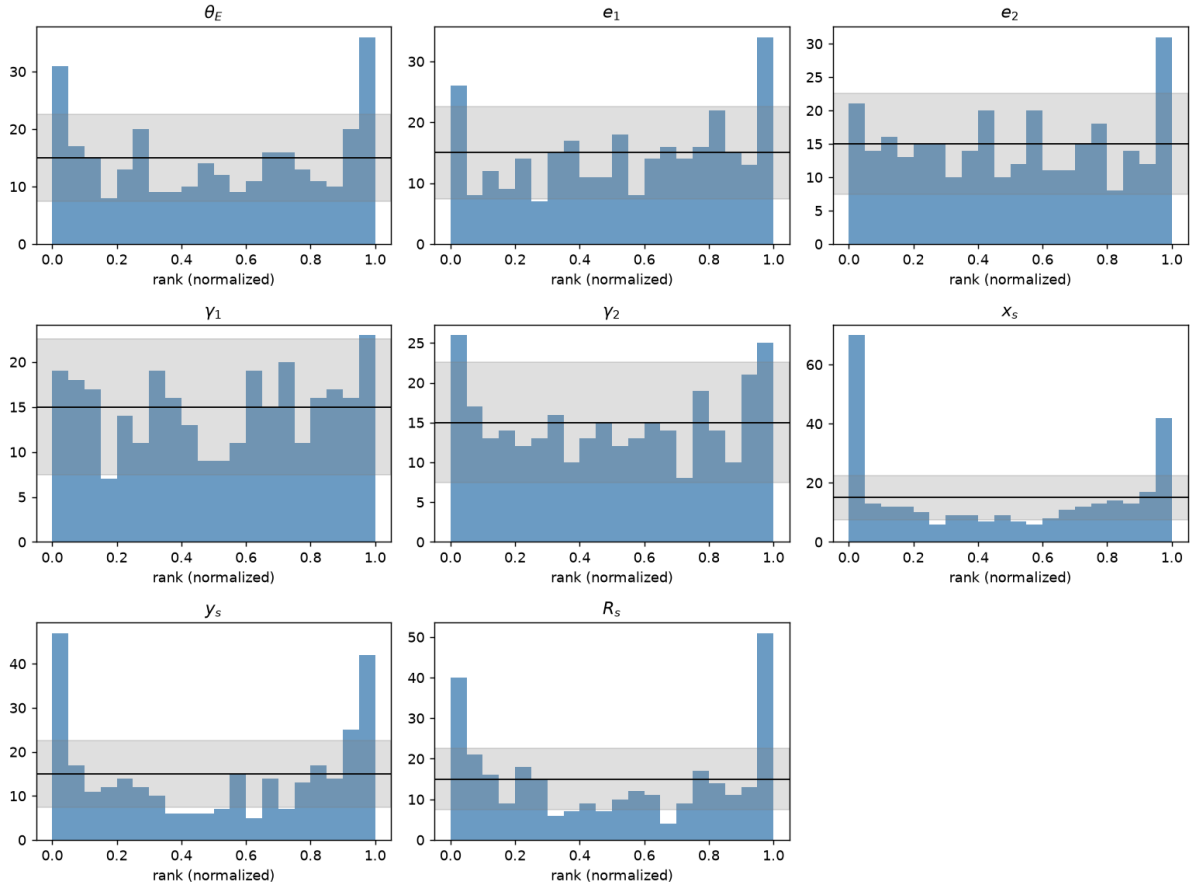


Figure 3: SBC rank histograms per parameter. Flat within the grey band indicates a calibrated posterior; a U-shape indicates overconfidence (posteriors too narrow).

The histograms reveal a consistent pattern:

- γ_1 and, to a lesser degree, e_2 are close to flat—the parameters the network constrains weakly inherit their width mostly from the prior, which is trivially calibrated.
- The well-recovered parameters (x_s, y_s, R_s, θ_E) show clear U-shapes with mass piled at ranks 0 and 1: the true value falls in the extreme tails of the posterior more often than

it should. The learned posteriors for these parameters are therefore *overconfident*—too narrow by a modest factor.

This overconfidence is a common failure mode of neural posterior estimators trained to convergence on limited data, and it is consistent with the overfitting seen in Figure 1: as the network memorizes training noise it becomes increasingly (and unjustifiably) certain. Practical mitigations are discussed in Section 5. The practical implication is that credible intervals for the well-constrained parameters should be inflated somewhat before being used for downstream science.

4.3 Posterior predictive check

As a complementary end-to-end test, we re-simulate images from posterior draws and compare them with the observed image (Figure 4). The posterior draws reproduce the observed arc configuration—ring radius, the positions of the three bright knots, and the overall azimuthal light distribution—demonstrating that the posterior concentrates on parameter values that regenerate the data. Differences between individual draws are small and concentrated in the brightness of the knots, visualizing the residual posterior uncertainty.

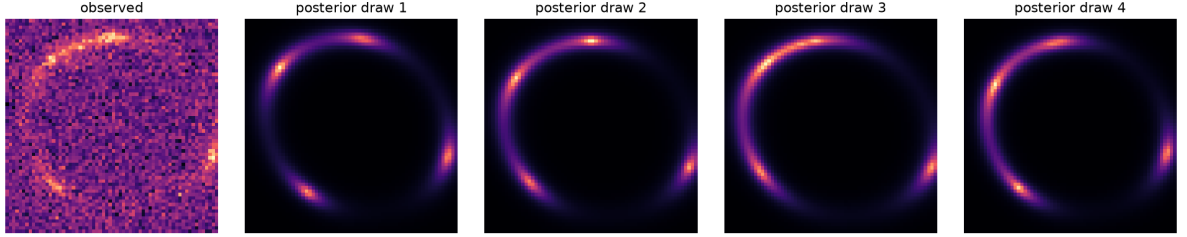


Figure 4: Posterior predictive check for one test system: the observed noisy image (left) and four noise-free re-simulations from independent posterior draws.

4.4 External convergence and the mass-sheet degeneracy

A uniform sheet of external convergence κ rescales the lens equation in a way that is almost perfectly absorbed by a rescaling of the (unobservable) source plane: image *morphology* alone cannot distinguish (κ, θ_E) from a suitably transformed pair. This is the classical *mass-sheet degeneracy*, and a sound inference pipeline must report it rather than hide it.

We repeat the full pipeline (generate, train, evaluate) with $\kappa \sim \mathcal{U}(0, 0.2)$ added as a ninth inferred parameter. The results show exactly the expected signature:

- **Minimal posterior contraction.** The prior standard deviation of κ is 0.058; the mean posterior standard deviation across test systems is 0.051, a contraction of only 12%. A single image leaves κ almost as uncertain as the prior—the network has correctly learned that this parameter is nearly unidentifiable.
- **Strong internal anti-correlation.** Within individual posteriors the mean correlation between κ and θ_E is -0.83 : the two parameters trade off against each other along the degeneracy direction, exactly as the lensing equations predict (a larger mass sheet requires a smaller intrinsic Einstein radius to produce the same ring).

That an off-the-shelf neural posterior estimator recovers this degeneracy structure—without being told about it—is a meaningful validation of the SBI approach: the network reports joint posterior *geometry*, not just marginal point estimates. In practice, breaking this degeneracy requires external information (stellar kinematics of the lens, time delays, or statistical priors on the line-of-sight environment), none of which is present in a single image.

5 Discussion and Limitations

Image signal-to-noise. The realized peak SNR (median ≈ 4) fell below the nominal 10–30 target, making this a stress test at the noisy end of realistic conditions. Raising the source amplitude or lowering the background RMS to hit the target range would likely improve ellipticity recovery substantially and shear recovery marginally; the mass-sheet degeneracy conclusions would be unchanged, as they are information-theoretic rather than noise-driven.

Overfitting and calibration. Training for the full 40 epochs past the validation plateau produced overconfident posteriors for the well-constrained parameters. Three inexpensive remedies, in increasing order of effort: early stopping on validation loss (epoch ~ 20); enlarging the training set (the simulator is cheap—50,000 simulations remain feasible, and now that training runs on GPU the enlarged dataset would still train quickly); and light regularization (dropout in the summary CNN or weight decay).

Model scope. The forward model fixes several quantities (lens centroid, source amplitude and profile shape, PSF) that would be uncertain for real observations, and simulates only the lensed source (no lens-galaxy light). Extending θ with these nuisance parameters, or marginalizing over them in the simulator, is straightforward within the same pipeline and is the natural next step toward application to survey data.

Computational profile. With GPU training (< 20 min) the pipeline bottleneck is now dataset simulation (~ 5 – 10 min, CPU-parallelized). Once trained, the amortized posterior costs milliseconds per system—the property that makes this approach attractive for the 10^4 – 10^5 lens samples expected from upcoming surveys.

6 Conclusion

We built and validated an end-to-end amortized SBI pipeline for strong-lens parameter inference. From single noisy 64×64 images, the BayesFlow posterior estimator recovers the Einstein radius, source position, and source size accurately ($R^2 = 0.83$ – 0.95), the lens ellipticity moderately, and correctly assigns wide uncertainties to external shear, which is nearly unidentifiable at the simulated noise level. Simulation-based calibration exposes mild overconfidence in the best-constrained parameters, attributable to overfitting and addressable with early stopping and more training data. When external convergence is included, the inferred posteriors exhibit the mass-sheet degeneracy exactly as theory predicts: negligible contraction of κ relative to the prior and a strong κ – θ_E anti-correlation (-0.83). The complete pipeline—simulation, training, and evaluation—runs in about half an hour on a single consumer GPU, after which inference on new systems is effectively instantaneous.

References

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